

EQUIVALENCES OF CATEGORIES RELEVANT TO COMBINATORICS AND REPRESENTATION THEORY

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1. INTRODUCTION

Our purpose is to understand and generalize the classification theorem for $\mathbf{FI}\sharp$ -modules appearing as Theorem 2.24 of [1]. Let $\mathfrak{S}$ be the groupoid of finite sets and bijective functions. Let $\mathbf{FI}\sharp$ denote the category of finite sets and injective partial functions. Let $\mathbf{Mod}^R$ denote the category of left modules over the ring R . Their Theorem 2.24 provides an equivalence between the category of functors $\mathfrak{S} \rightarrow \mathbf{Mod}^R$ and the category of functors $\mathbf{FI}\sharp \rightarrow \mathbf{Mod}^R$. The result gives a new viewpoint on representations of the symmetric groups, also a new viewpoint on Joyal species [7, 8].

2. THE KERNEL MODULE

Let us begin with any category $\mathcal{A}$ with pullback of monomorphisms along arbitrary morphisms. Recall that a morphism $s : B \rightarrow A$ is called a *strong epimorphism* when, for all monomorphisms $m : X \rightarrow Y$, the following square is a pullback.

$$\begin{array}{ccc}
 \mathcal{A}(A, X) & \xrightarrow{\mathcal{A}(e, X)} & \mathcal{A}(B, X) \\
 \mathcal{A}(A, m) \downarrow & & \downarrow \mathcal{A}(B, m) \\
 \mathcal{A}(A, Y) & \xrightarrow{\mathcal{A}(e, Y)} & \mathcal{A}(B, Y)
 \end{array} \tag{2.1}$$

Let $\mathcal{A}_m$ denote the category whose objects are all those of $\mathcal{A}$ and whose morphisms are only the monomorphisms in $\mathcal{A}$. Let $\mathcal{A}_{se}$ denote the category whose objects are

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all those of $\mathcal{A}$ and whose morphisms are only the strong epimorphisms in $\mathcal{A}$. By properties of pullback, strong epimorphisms do compose.

A span $f = (X \xleftarrow{f_0} U \xrightarrow{f_1} Y)$ is called a *partial map* $f : X \longrightarrow Y$ when f_0 is a monomorphism. Let $\text{Par}\mathcal{A}$ denote the category whose objects are all those of $\mathcal{A}$ and whose morphisms are isomorphism classes $[f]$ of partial maps. Composition is that of spans: that is, by pullback. There are functors $(-)^* : \mathcal{A}_m^{\text{op}} \longrightarrow \text{Par}\mathcal{A}$ and $(-)_* : \mathcal{A} \longrightarrow \text{Par}\mathcal{A}$ which are the identity on objects. For a monomorphism $m : U \longrightarrow X$ in $\mathcal{A}$, define $m^* = [m, U, 1_U] : X \longrightarrow U$. For any $f : X \longrightarrow Y$ in $\mathcal{A}$, define $f_* = [1_X, X, f] : X \longrightarrow Y$. Every partial map $f = (X \xleftarrow{f_0} U \xrightarrow{f_1} Y)$ has

$$[f] = f_{1*} \circ f_0^* .$$

For every pullback

$$\begin{array}{ccc} P & \xrightarrow{g} & B \\ n \downarrow & & \downarrow m \\ A & \xrightarrow{f} & C \end{array} \quad (2.2)$$

in $\mathcal{A}$ with m a monomorphism, the square

$$\begin{array}{ccc} P & \xrightarrow{g_*} & B \\ n^* \uparrow & & \uparrow m^* \\ A & \xrightarrow{f_*} & C \end{array} \quad (2.3)$$

commutes in $\text{Par}\mathcal{A}$.

There is analogue, for partial maps in place of general spans, of the result in [13] about Mackey functors. There is an equivalence between functors $T : \text{Par}\mathcal{A} \longrightarrow \mathcal{X}$ and pairs (T^*, T_*) of functors $T^* : \mathcal{A}_m^{\text{op}} \longrightarrow \mathcal{X}$ and $(-)^* : \mathcal{A} \longrightarrow \mathcal{X}$ which agree on objects and satisfy the Bénabou-Roubaud-Chevalley-Beck condition on pullbacks of the form (2.2). Indeed $T^* = T \circ (-)^*$ and $T_* = T \circ (-)_*$.

Let Set denote the category of sets and functions. Here the monomorphisms are the injective functions and the strong epimorphisms are the surjective functions.

Notice that, if all morphisms in $\mathcal{A}$ are monomorphisms (such as in the category of finite sets and injective functions), then the strong epimorphisms are the invertible morphisms.

There is a “kernel” functor

$$M : \mathcal{A}_{\text{se}}^{\text{op}} \times \text{Par}\mathcal{A} \longrightarrow \text{ParSet} \quad (2.4)$$

defined on objects by

$$M(A, X) = \mathcal{A}(A, X) .$$

It is defined on morphisms by

$$\begin{array}{ccc}
 M(A, X) = \mathcal{A}(A, X) & & \\
 \downarrow M(s, f) & \swarrow \mathcal{A}(A, f_0) & \\
 & \mathcal{A}(A, U) & \\
 & \swarrow \mathcal{A}(s, f_1) & \\
 M(B, Y) = \mathcal{A}(B, Y) & &
 \end{array} \quad (2.5)$$

where $s : B \rightarrow A$ is a strong epimorphism and $[f] = [f_0, U, f_1] : X \rightarrow Y$ is a partial map in $\mathcal{A}$. To see that composition is preserved, take a strong epimorphism $t : C \rightarrow B$ and partial map $[g] = [g_0, V, g_1] : Y \rightarrow Z$, and form the pullback P of f_1 and g_0 as in the diamond in the diagram:

$$\begin{array}{ccccc}
 & & P & & \\
 & \swarrow h_0 & \downarrow p_0 & \downarrow p_1 & \searrow h_1 \\
 & U & & V & \\
 \swarrow f_0 & & \downarrow f_1 & & \swarrow g_0 \\
 X & & Y & & Z
 \end{array} \quad (2.6)$$

Then we have

$$\begin{array}{ccccc}
 & & \mathcal{A}(A, P) & & \\
 & \swarrow \mathcal{A}(A, h_0) & \downarrow \mathcal{A}(A, p_0) & \downarrow \mathcal{A}(s, p_1) & \searrow \mathcal{A}(st, h_1) \\
 & \mathcal{A}(A, U) & & \mathcal{A}(B, V) & \\
 \swarrow \mathcal{A}(A, f_0) & & \downarrow \mathcal{A}(s, f_1) & & \swarrow \mathcal{A}(B, g_0) \\
 \mathcal{A}(A, X) & & \mathcal{A}(B, Y) & & \mathcal{A}(C, Z)
 \end{array} \quad (2.7)$$

where the diamond is a pullback obtained as a pasting of two pullbacks, one pullback is (2.5) for the strong epimorphism s and the monomorphism p_0 , and the other is the image under $\mathcal{A}(B, -)$ of the pullback (2.6). That is,

$$M(t, [g]) \circ M(s, [f]) = [\mathcal{A}(A, h_0), \mathcal{A}(A, P), \mathcal{A}(st, h_1)] = M(st, [g] \circ [f]) .$$

So M preserves composition. That M preserves identity morphisms is clear.

We write

$$\overline{M} : \mathcal{A}_{\text{se}}^{\text{op}} \rightarrow [\text{Par}\mathcal{A}, \text{ParSet}]$$

for the functor corresponding to (2.4) using cartesian closedness of CAT. On objects, $\overline{M}(A) = M(A, -)$.

Note that, in ParSet, we can identify morphisms $[f] = [f_0, U, f_1] : X \rightarrow Y$ with their canonical unique representative f of the form (f_0, U, f_1) where $U \subseteq X$ (called the *domain of definition* of f) and f_0 is the inclusion.

There is a well known equivalence of categories

$$1 + - : \text{ParSet} \simeq 1/\text{Set} \quad (2.8)$$

where $1/\text{Set}$ is the category of pointed sets and point-preserving functions. The equivalence $1 + -$ takes a set X to the set $1 + X$ which is X with a disjointly added point

0 and takes a partial function $f : X \rightarrow Y$ to the function $1 + f : 1 + X \rightarrow 1 + Y$ defined by

$$(1 + f)(x) = \begin{cases} 0 & \text{for } x \notin U \\ f(x) & \text{for } x \in U \end{cases},$$

where U is the domain of definition of f .

The equivalence (2.8) makes it clear that ParSet is complete and cocomplete. For later purposes, we remind the reader that coproducts in $1/\text{Set}$ are formed by taking the disjoint union and then identifying all the distinguished points 0. Since $[\mathcal{A}_{\text{se}}, 1/\text{Set}]$ is the free cocompletion of the free $1/\text{Set}$ -enriched category on $\mathcal{A}_{\text{se}}^{\text{op}}$ and since $[\text{Par}\mathcal{A}, 1/\text{Set}]$ is a cocomplete $1/\text{Set}$ -enriched category, the functor $\overline{M}$ and the equivalence (2.8) determine a colimit-preserving (pointed) functor

$$\widehat{(-)} : [\mathcal{A}_{\text{se}}, 1/\text{Set}] \rightarrow [\text{Par}\mathcal{A}, 1/\text{Set}]. \quad (2.9)$$

We now look in more detail at (2.9) and its right adjoint

$$\widetilde{(-)} : [\text{Par}\mathcal{A}, 1/\text{Set}] \rightarrow [\mathcal{A}_{\text{se}}, 1/\text{Set}]. \quad (2.10)$$

3. THE TRANSFORM ADJUNCTION

We continue with a category $\mathcal{A}$ as in Section 2. Now we further assume that each morphism f can be factored as $f = me$ where m is a monomorphism and e is a strong epimorphism. It follows that the factorization is unique up to isomorphism.

To simplify the exposition, we assume there are special monomorphisms $i_U : U \rightarrow X$, called *inclusions*, which form a partially ordered subcategory of $\mathcal{A}$, and each morphism $f : X \rightarrow Y$ in $\mathcal{A}$ has a unique factorization $f = i_V e$ where $i_V : V \rightarrow Y$ is an inclusion and e is a strong epimorphism. We may also assume that, if g and $g \circ f$ are inclusions then f is. We write $U \subseteq X$ when there is an inclusion $i_U : U \rightarrow X$; the inclusion is necessarily unique. It follows that we can choose the pullback of an inclusion along any morphism to be an inclusion. Also, the pullback of two inclusions gives a square with all four morphisms inclusions: for $S \subseteq A$ and $S_1 \subseteq A$, we write $S \cap S_1$ for the pullback of the inclusions.

Morphisms of $\text{Par}\mathcal{A}$ can be written as $(U, f) : X \rightarrow Y$ where $U \subseteq X$ and $f : U \rightarrow Y$.

We shall define a functor

$$\widehat{(-)} : [\mathcal{A}_{\text{se}}, 1/\text{Set}] \rightarrow [\text{Par}\mathcal{A}, 1/\text{Set}] \quad (3.11)$$

and show that it agrees with (2.9).

For $H : \mathcal{A}_{\text{se}} \rightarrow 1/\text{Set}$ and $X \in \text{Par}\mathcal{A}$, define

$$\widehat{H}X = \sum_{A \subseteq X} HA \quad (3.12)$$

where the sum is the coproduct of pointed sets. For each $(U, f) : X \rightarrow Y$ in $\text{Par}\mathcal{A}$, define

$$\widehat{H}(U, f) : \sum_{A \subseteq X} HA \rightarrow \sum_{B \subseteq Y} HB \quad (3.13)$$

by taking its composite $\widehat{H}(U, f) \circ \text{in}_A$ with the injection at $A \subseteq X$ to be 0 unless $A \subseteq U$, in which case it is the composite $\text{in}_B \circ Hs$ where $f \circ i_U = i_B \circ s$ with s a strong epimorphism.

Proposition 3.1. *Under the given conditions on $\mathcal{A}$, the functors (2.9) and (3.11) are isomorphic.*

Proof. Since (3.11) preserves colimits, it suffices to prove the functors agree on objects of the form $H_K = 1 + \mathcal{A}_{\text{se}}(K, -)$. We have

$$\widehat{H}_K X = 1 + \sum_{A \subseteq X} \mathcal{A}_{\text{se}}(K, A) \cong 1 + \mathcal{A}(K, A) \cong 1 + (\overline{M}K)X ,$$

showing that (2.9) and (3.11) agree on such objects. We leave the check on morphisms to the reader. $\square$

Suppose we have a functor $T : \text{Par}\mathcal{A} \rightarrow 1/\text{Set}$. Let us say $x \in TX$ is *supported* when, for all monomorphisms $m : U \rightarrow X$, if $(Tm^*)x \neq 0$ then m is invertible. Define

$$\widetilde{TX} = \{x \in TX : x \text{ is supported}\} . \quad (3.14)$$

Notice that 0 is always supported. Also, it suffices to test supportedness using inclusions for m .

Lemma 3.2. *For a strong epimorphism $s : X \rightarrow Y$, the function $Ts_* : TX \rightarrow TY$ restricts to give a function $\widetilde{Ts} : \widetilde{TX} \rightarrow \widetilde{TY}$.*

Proof. We need to see that if $x \in TX$ is supported then so is $(Ts_*)x \in TY$. Take a monomorphism $n : V \rightarrow Y$ such that $(Tn^*)(Ts_*)x \neq 0$. Form the pullback

$$\begin{array}{ccc} U & \xrightarrow{q} & V \\ m \downarrow & & \downarrow n \\ X & \xrightarrow{s} & Y \end{array} \quad (3.15)$$

in $\mathcal{A}$. Using $(Tn^*)(Ts_*) = (Tq_*)(Tm^*)$, we see that $(Tm^*)x \neq 0$. Since x is supported, m is invertible. So nq and hence n are strong epimorphisms. So n is invertible. This shows that $(Ts_*)x \in \widetilde{TY}$. $\square$

Theorem 3.3. *The functor (3.11) has a right adjoint taking the functor $T : \text{Par}\mathcal{A} \rightarrow 1/\text{Set}$ to $\widetilde{T}$. The unit of the adjunction is invertible; in other words, (3.11) is fully faithful. The counit of the adjunction is a monomorphism.*

Proof. We must construct a natural bijection

$$[\text{Par}\mathcal{A}, 1/\text{Set}](\widehat{H}, T) \cong [\mathcal{A}_{\text{se}}, 1/\text{Set}](H, \widetilde{T}) . \quad (3.16)$$

Consider a natural transformation $\theta : \widehat{H} \Rightarrow T$. Let $\theta_{X,A} : HA \rightarrow TX$ denote the restriction of θ_X along the $(A \subseteq X)$ -component of the injection. Observe that $\theta_{A,A} : HA \rightarrow TA$ lands in $\widetilde{TA}$. For, suppose we have a $U \subseteq A$ with $(Ti_U^*)\theta_{A,A}a \neq 0$ invertible for some $a \in HA$. Then, by naturality, $\theta_{A,U}(\widehat{H}i_U^*)a \neq 0$. Of course, then $(\widehat{H}i_U^*)a \neq 0$, and by definition of $\widehat{H}$ on morphisms, $A \subseteq U$. So $U = A$, as required.

Now we can define $\phi_A : HA \rightarrow \tilde{T}A$ by $\phi_A a = \theta_{A,A} a$. We shall show that these morphisms are the components of a natural transformation $\phi : H \Rightarrow \tilde{T}$. Take any strong epimorphism $s : A \rightarrow B$. Then

$$(\tilde{T}s)\phi_A a = (Ts_*)\theta_{A,A} a = \theta_B(\hat{H}s_*)\text{in}_A a = \theta_{B,B}(Hs)a = \phi_B(Hs)a$$

using the naturality of θ . So we have a natural function $\theta \mapsto \phi$ for (3.16).

For the inverse bijection, take a natural transformation $\phi : H \rightarrow \tilde{T}$ and define $\theta_X : \hat{H}X \rightarrow TX$ by

$$\theta_{X,A} a = (Ti_{A*})\phi_A a.$$

We prove naturality of θ in two steps, first with respect to morphisms i_U^* for $U \subseteq X$ and then with respect to f_* for $f : X \rightarrow Y$ in $\mathcal{A}$. For the first, form the pullback span $(u, W, v) : A \rightarrow V$ of the pair $i_A : A \rightarrow X$ and $i_U : U \rightarrow X$. Then

$$(Ti_U^*)\theta_{X,A} a = (Ti_U^*)(Ti_{A*})\phi_A a = (Tv_*)(Tu^*)\phi_A a = T(v_*u^*)\phi_A a$$

which is 0 unless $(Tu^*)\phi_A a \neq 0$. However, $\phi_A a$ is supported, so u is invertible. On the other hand, $\theta_U(\hat{H}i_U^*)\text{in}_A a = 0$ unless $A \subseteq V$ with inclusion j , say, in which case

$$\theta_U(\hat{H}i_U^*)\text{in}_A a = \theta_{U,A} a = (Tj_*)\phi_A a = T(v_*u^*)\phi_A a.$$

Now we turn to naturality of ϕ with respect to f_* . This requires factoring $fi_A = i_B s$ with s strong epimorphic. Then

$$\begin{aligned} \theta_Y(\hat{H}f_*)a &= \theta_Y(Hs)a = (Ti_{B*})\phi_B(Hs)a = (Ti_{B*})(\tilde{T}s)\phi_A a \\ &= T(i_{B*}s_*)\phi_A a = T(f_*i_{A*})\phi_A a = (Tf_*)\theta_X a. \end{aligned}$$

It is easy to see that $\theta \mapsto \phi$ and $\phi \mapsto \theta$ are mutually inverse.

The component of the unit $\eta_H : H \Rightarrow \hat{H}$ at A is just the injection $\text{in}_A : HA \rightarrow \hat{H}A$ seen as landing in the supported elements; so it is injective. Take any supported $\text{in}_U x \in \hat{H}A$. So $x \in HU$. If $x = 0$ then it is in the image of $\eta_H A$. If $x \neq 0$ then $(\hat{H}i_U^*)x = x \neq 0$, and, by supportedness, $U = A$; so $\text{in}_U x = \text{in}_A x$ is in the image of $\eta_H A$. So η_H is invertible.

The component of the counit $\varepsilon_T : \hat{T} \rightarrow T$ at X takes $a \in \tilde{T}A$ to $(Ti_{A*})a \in TX$ for $A \subseteq X$. If $(Ti_{A*})a = 0$ then $(Ti_A^*)(Ti_{A*})a = 0$; but $i_A^*i_{A*} = 1_A$, so $a = 0$. If $(Ti_{A*})a = (Ti_{B*})b \neq 0$ with $a \in \tilde{T}A$, $b \in \tilde{T}B$, then $T(i_B^*i_{A*})a = T(i_B^*i_{B*})b = b$. Form the pullback (u, P, v) of i_A and i_B . Then we have $T(u_*v^*)a = T(i_B^*i_{A*})a = b \neq 0$. So $T(v^*)a \neq 0$. Since a is supported, v is invertible. It follows that $A \subseteq B$. Symmetrically, $B \subseteq A$. So $A = B$ and $a = b$. This proves injectivity of $\varepsilon_T X$. So ε_T is a monomorphism. $\square$

Remark 3.4. It seems that this result carries through for proper factorization systems [6] on $\mathcal{A}$ other than that of strong epimorphisms and monomorphisms.

Remark 3.5. The essential image of (3.11) consists of those T with the property that each $x \in TX$ is of the form $x = T(i_{U*})u$ for some $U \subseteq X$ and supported $u \in TU$.

4. THE MONOMORPHISM CASE WITH ADDITIVITY

There is no problem replacing $1/\text{Set}$ in Section 3 by the category Mod^R . The same formula (3.12) provides a functor

$$\widehat{(-)} : [\mathcal{A}_{\text{se}}, \text{Mod}^R] \longrightarrow [\text{Par}\mathcal{A}, \text{Mod}^R] . \quad (4.17)$$

Similarly, the formula (3.14) gives the value at T of a right adjoint

$$\widetilde{(-)} : [\text{Par}\mathcal{A}, \text{Mod}^R] \longrightarrow [\mathcal{A}_{\text{se}}, \text{Mod}^R] . \quad (4.18)$$

to (4.17). Notice that $\widetilde{TX}$ is a submodule of TX : supported elements are closed under addition and scalar multiple (if $(Tm^*)(x + y) = (Tm^*)x + (Tm^*)y \neq 0$ then either $(Tm^*)x \neq 0$ or $(Tm^*)y \neq 0$; and, if $(Tm^*)(rx) = r(Tm^*)x \neq 0$ then $(Tm^*)x \neq 0$).

The proof of Theorem 3.3 also proves:

Theorem 4.1. *The functor (4.17) has the right adjoint (4.18). The unit of the adjunction is invertible; in other words, (4.17) is fully faithful. The counit of the adjunction is a monomorphism.*

With some further assumptions on $\mathcal{A}$, we shall show that the adjunction of Theorem 4.1 is an equivalence.

Here are our further assumptions on $\mathcal{A}$:

- (a) all morphisms are monomorphisms (so that strong epimorphisms are invertible);
- (b) there is a natural-number-valued function $\# : \text{ob}\mathcal{A} \longrightarrow \mathbb{N}$ defined on the objects of $\mathcal{A}$;
- (c) if $f : A \longrightarrow B$ in $\mathcal{A}$ then $\#A \leq \#B$;
- (d) for each natural number n and object $A \in \mathcal{A}$, the set $\binom{A}{n} = \{U \subseteq A : \#U = n\}$ is finite;
- (e) if $U \subseteq A$ and $\#U = \#A$ then $U = A$.

Each $U \subseteq A$ in $\mathcal{A}$ determines an idempotent $(U, i_U) : A \longrightarrow A$ in $\text{Par}\mathcal{A}$. Applying $T : \text{Par}\mathcal{A} \longrightarrow \text{Mod}^R$, we obtain an idempotent

$$e_A^U = T(U, i_U) : TA \longrightarrow TA$$

on the R -module TA .

Fix a natural number n . Assume $TA = 0$ for $\#A < n$. In this case, if $U \neq V \in \binom{A}{n}$, then the idempotents e_A^U and e_A^V are orthogonal; that is, their composite in either order is 0. This is because, using (e) above, $\#(U \cup V) < n$, so $T(U \cap V) = 0$. Hence, we obtain an idempotent $e_A^n : TA \longrightarrow TA$ defined by

$$e_A^n = \sum_{U \in \binom{A}{n}} e_A^U . \quad (4.19)$$

Proposition 4.2. *Under the current assumptions on $\mathcal{A}$ and T , the idempotents (4.19), for $A \in \text{Par}\mathcal{A}$, are the components of a natural transformation $e^n : T \Longrightarrow T$.*

Proof. First we look at naturality with respect to morphisms $(U, 1_U) : A \longrightarrow U$. We have

$$T(U, 1_U)e_A^n = \sum_{V \in \binom{A}{n}} T(V \cap U, i_{V \cap U}) .$$

However, to have $T(V \cap U) \neq 0$ we must have $V \subseteq U$ and $V \cap U = V$. So

$$T(U, 1_U) e_A^n = \sum_{V \in \binom{U}{n}} T(V, i_V) = e_U^n T(U, 1_U) .$$

Now we look at naturality with respect to morphisms $(A, f) : A \longrightarrow B$. We have

$$T(A, f) e_A^n = \sum_{U \in \binom{A}{n}} T(U, f i_U) ,$$

and

$$e_B^n T(A, f) = \sum_{V \in \binom{B}{n}} T(f^* V, i_V \bar{f}) ,$$

where $\bar{f} : f^* V \longrightarrow V$ is the pullback of f along i_V . The direct image $f_* U \subseteq B$ of $U \subseteq A$ is obtained by factoring $f i_U = i_{f_* U} s$ where s is a strong epimorphism. Since f is a monomorphism, we have $U = f^* f_* U$ and $f_* U \in \binom{B}{n}$ for $U \in \binom{A}{n}$. Also we have $f_* f^* V \subseteq V$ so $\# f^* V \leq n$ for $V \in \binom{B}{n}$. However, if $\# f^* V < n$, $T(f^* V, i_V \bar{f}) = 0$. So we only need to take the second sum over the $V = f_* U$, showing the two sums equal. $\square$

For any $T : \text{Par}\mathcal{A} \longrightarrow \text{Mod}^R$, we define a functor $H_T^n : \mathcal{A}_{\text{se}} \longrightarrow \text{Mod}^R$ by the truncation

$$H_T^n A = \begin{cases} 0 & \text{for } \#A > n \\ TA & \text{for } \#A \leq n \end{cases} , \quad (4.20)$$

and $H_T^n f = T(A, f)$ for $f : A \longrightarrow B$ in $\mathcal{A}_{\text{se}}$.

With our assumption on T that it is 0 on objects A with $\#A < n$, we see that

$$\widehat{H_T^n} X = \sum_{A \subseteq X} H_T^n A = \sum_{A \in \binom{X}{n}} TA = \sum_{A \in \binom{X}{n}} e_X^A TX = (e^n T) X .$$

So the image of the idempotent e^n on T is $\widehat{H_T^n}$. Thus we have the direct sum decomposition

$$T = \widehat{H_T^n} \oplus T_{n+1} \quad (4.21)$$

in $[\text{Par}\mathcal{A}, \text{Mod}^R]$ where $T_{n+1} = (1_T - e^n)T$ is 0 on objects A with $\#A < n+1$ (since $\binom{A}{n} = \{A\}$ when $\#A = n$).

Now take any $T : \text{Par}\mathcal{A} \longrightarrow \text{Mod}^R$. We define a sequence of functors $T_n : \text{Par}\mathcal{A} \longrightarrow \text{Mod}^R$ recursively by: $T_0 = T$ and

$$T_n = \widehat{H_{T_n}^n} \oplus T_{n+1} .$$

Since (4.17) preserves colimits, it follows that

$$T \cong \widehat{\bigoplus_{n \geq 0} H_{T_n}^n} . \quad (4.22)$$

This proves (4.17) is essentially surjective.

Theorem 4.3. *Under the conditions (a)–(e) on $\mathcal{A}$, the functor $\widehat{(-)}$ (4.17) is an equivalence.*

A *(set) species* in the sense of Joyal [7] is a functor $H : \mathfrak{S} \longrightarrow \text{Set}$ where $\mathfrak{S}$ is the category of sets and bijective functions. A *pointed-set species* is a functor $H : \mathfrak{S} \longrightarrow 1/\text{Set}$. An *R -module species* is a functor $H : \mathfrak{S} \longrightarrow \text{Mod}^R$; the case where R is a field is the basic situation of [8].

Following [1], we write FI for the category of finite sets and injective functions. Assumptions (a)–(e) all hold for $\mathcal{A} = \text{FI}$ with $\#$ denoting cardinality. We then see that $\mathcal{A}_m = \text{FI}$ and $\mathcal{A}_{\text{se}} = \mathfrak{S}$, while $\text{Par}\mathcal{A} = \text{FI}_{\#}$, the category of injective partial functions.

Corollary 4.4. [1] *The functor*

$$\widehat{(-)} : [\mathfrak{S}, \text{Mod}^R] \longrightarrow [\text{FI}_{\#}, \text{Mod}^R]$$

is an equivalence of categories.

Another example relevant to [11] might be the category $\mathcal{A} = \text{FIVect}_{\mathbb{F}_q}$ of finite vector spaces over the field $\mathbb{F}_q$ of cardinality q (a prime power) and injective linear functions. Here $\#$ is dimension. Let $\mathcal{A}_{\text{se}} = \mathfrak{S}\mathfrak{I}_q$ be the category of finite $\mathbb{F}_q$ -vector spaces and linear isomorphisms. Then $\text{Par}\mathcal{A} = \text{FI}_{\#q}$ is the category of finite $\mathbb{F}_q$ -vector spaces and injective partial linear functions.

Corollary 4.5. *The functor*

$$\widehat{(-)} : [\mathfrak{S}\mathfrak{I}_q, \text{Mod}^R] \longrightarrow [\text{FI}_{\#q}, \text{Mod}^R]$$

is an equivalence of categories.

5. APPENDIX: REMARKS ON THE DOLD-PUPPE-KAN EQUIVALENCE

Let $\mathbb{D}$ be the additive category whose objects are the natural numbers, whose homs are

$$\mathbb{D}(m, n) = \begin{cases} \mathbb{Z} & \text{for } n = m, m+1, \\ 0 & \text{otherwise} \end{cases},$$

and whose composition

$$\mathbb{D}(n, p) \otimes \mathbb{D}(m, n) \longrightarrow \mathbb{D}(m, p)$$

is zero unless all three homs are $\mathbb{Z}$ in which case it is the canonical isomorphism $\mathbb{Z} \otimes \mathbb{Z} \cong \mathbb{Z}$. We write $\partial_n : n \longrightarrow n+1$ and $1_n : n \longrightarrow n$ for $1 \in \mathbb{Z}$.

For any additive category $\mathcal{V}$, the additive functor category $[\mathbb{D}^{\text{op}}, \mathcal{V}]$ is the category of differential non-negatively graded $\mathcal{V}$ -objects (or $\mathcal{V}$ -chain complexes).

Let Δ_{inj} denote the monoidal subcategory of the algebraic simplicial category Δ whose objects are all the natural numbers and whose morphisms $m \longrightarrow n$ are order-preserving injective functions

$$\mu : \{0, 1, \dots, m-1\} \longrightarrow \{0, 1, \dots, n-1\}.$$

This is the PRO for coaugmented objects: that is, objects equipped with a morphism from the tensor unit. Let $K : \Delta_{\text{inj}} \longrightarrow \Delta$ denote the inclusion.

There is a functor $J : \Delta_{\text{inj}} \longrightarrow \mathbb{D}$ which is the identity on objects and takes the injection $\partial_i : n-1 \longrightarrow n$ to 0 for $i < n$ and to ∂_n for $i = n$.

At the heart of the Dold-Puppe-Kan Theorem is the span of functors

$$\mathbb{D} \xleftarrow{J} \Delta_{\text{inj}} \xrightarrow{K} \Delta.$$

For any simplicial object $X : \Delta \rightarrow \mathcal{V}$, consider the chain complex $N(X)$ defined as the right Kan extension of XK^{op} along J^{op} , which exists under a mild completeness condition on $\mathcal{V}$.

$$\begin{array}{ccc}
 \Delta_{\text{inj}}^{\text{op}} & \xrightarrow{J^{\text{op}}} & \mathbb{D}^{\text{op}} \\
 K^{\text{op}} \downarrow & \xleftarrow{\rho} & \downarrow N(X) \\
 \Delta^{\text{op}} & \xrightarrow{X} & \mathcal{V}
 \end{array} \tag{5.23}$$

We have the weighted limit formula

$$N(X)_n = \lim(\mathbb{D}(J-, n), XK^{\text{op}}) .$$

The weight functor $\mathbb{D}(J-, n) : \Delta_{\text{inj}}^{\text{op}} \rightarrow \text{Ab}$ into the category Ab of abelian groups takes all objects to 0 except that $n-1$ and n go to $\mathbb{Z}$, and takes the morphisms $\partial_i : n-1 \rightarrow n$ to 0 for $0 \leq i < n$ and to the identity abelian group morphism $\mathbb{Z} \rightarrow \mathbb{Z}$ for $i = n$. The natural transformations $\mathbb{D}(J-, n) \Rightarrow SK^{\text{op}}$ for a simplicial abelian group S are clearly in bijection with elements $x \in S_n$ such that $d_i(x) = 0$ for $0 \leq i < n$. It follows that $N(X)_n$ is the intersection of the kernels of the $d_i : X_n \rightarrow X_{n-1}$ for $0 \leq i < n$. Also, $d_n : N(X)_n \rightarrow N(X)_{n-1}$ is induced by $d_n : X_n \rightarrow X_{n-1}$, making $N(X)$ a chain complex in $\mathcal{V}$.

We have the composite adjunction

$$\begin{array}{ccccc}
 [\mathbb{D}^{\text{op}}, \mathcal{V}] & \xrightarrow{\text{Res}_{J^{\text{op}}}} & [\Delta_{\text{inj}}^{\text{op}}, \mathcal{V}] & \xrightarrow{\text{Lan}_{K^{\text{op}}}} & [\Delta^{\text{op}}, \mathcal{V}] \\
 & \xleftarrow{\perp} & & \xleftarrow{\perp} & \\
 & \text{Ran}_{J^{\text{op}}} & & \text{Res}_{K^{\text{op}}} &
 \end{array}$$

where Res_H denotes restriction along H and its adjoints are left and right Kan extension along H . In our cases of H , the Kan extensions exist when $\mathcal{V}$ is finitely complete (and so has a bit of cocompleteness since it has finite direct sums).

Above we have identified the composite right-adjoint functor $\text{Ran}_{J^{\text{op}}}\text{Res}_{K^{\text{op}}}$ as the normalization functor N from simplicial objects to chain complexes. It is this functor that is proved an equivalence by Dold-Puppe-Kan [4, 5, 12].

Since K is strong monoidal, it follows from [2] that $\text{Lan}_{K^{\text{op}}}$ is strong monoidal for the convolution monoidal structures on $[\Delta^{\text{op}}, \mathcal{V}]$ and $[\Delta_{\text{inj}}^{\text{op}}, \mathcal{V}]$.

Although I have not completed the calculation, it seems that $\text{Res}_{J^{\text{op}}}$ is also strong monoidal for convolution monoidal structures on $[\Delta_{\text{inj}}^{\text{op}}, \mathcal{V}]$ and the usual tensor product on $[\mathbb{D}^{\text{op}}, \mathcal{V}]$. Since $\text{Res}_{J^{\text{op}}}$ is colimit preserving, we only need check the result on representables.

This would imply that $N : [\Delta^{\text{op}}, \mathcal{V}] \rightarrow [\mathbb{D}^{\text{op}}, \mathcal{V}]$ is a monoidal equivalence.

The monoidal structure on Δ is not symmetric and so neither is the monoidal structure on $[\Delta^{\text{op}}, \text{Set}]$. As often happens, when we move to the additive context, a symmetry can be found: just transport the usual symmetry on chain complexes across the equivalence N .

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